

BANK LENDING ANALYSIS EXECUTIVE REPORT

EXECUTIVE SUMMARY

The primary goal of this project is to improve lending decisions by moving away from guesswork and using actual data to see if current strategies are working. By establishing a clear way to see how loans are performing, the bank can make better, safer choices for the future. We have identified the health of our money by monitoring vital signs like cash flow and application volume, allowing us to distinguish between profitable good loans and high risk bad loans. This data-driven approach ensures the bank stays profitable while keeping loans affordable for our customers.

KEY INSIGHTS

- **Total Loan Applications:** The bank received a total of 38.6K loan applications showing a high demand for credit products.
- **Capital Flow:** The bank handed out a total of 435.8M to customers while successfully recovering 473.1M in total cash flow.
- **Good Loan Ratio:** 86.1% of the portfolio consists of good loans where customers are paying on time or have fully paid.
- **Bad Loan Risk:** 13.9% of the business is categorized as bad loans or charged off which helps the bank manage risk by identifying unlikely payments.
- **Monthly Growth Trends:** The month over month tracking shows a 6.9% increase in applications and a 13.0% increase in the total funded amount.
- **Borrower Purpose:** Data shows that the most common reason for borrowing is debt consolidation followed by credit card refinancing and home repairs.
- **Average Interest Rate:** The bank charges an average interest rate of 12.0% across the entire loan portfolio.
- **Debt to Income Ratio:** The average debt to income ratio for borrowers is 13.3% which ensures the loans stay affordable.
- **Human Factors:** Trends show that job stability and employment length are key indicators of who is borrowing and their ability to pay back.

KEY RECOMMENDATIONS

- **Update Lending Criteria:** The bank should use the data from the 13.9% bad loan segment to tighten approval rules for applicants with high debt to income ratios.
- **Targeted Financial Products:** Since debt consolidation is the primary reason for borrowing we should create specialized loan packages for this specific customer need.
- **Regional Investment:** Marketing efforts should be increased in high activity states like California and Texas where the data shows the most significant lending volume.
- **Employment Stability Incentives:** The bank should offer better terms or lower interest rates to borrowers with long term job stability as they represent a safer investment.
- **Proactive Collection Strategy:** Management should use the monthly tracking of vital signs to increase collection efforts immediately if the amount received starts to shrink.

IMPACT

- **Risk Reduction:** Implementing stricter rules for high risk segments will directly lower the 13.9% bad loan rate and protect bank capital.
- **Increased Profitability:** Focusing on high volume states and stable employees ensures a higher percentage of interest recovery and better cash flow.
- **Customer Retention:** Offering specialized products for debt consolidation builds stronger relationships with borrowers by meeting their most frequent needs.
- **Operational Efficiency:** Moving from guesswork to data-driven tracking allows the bank to adjust strategies quickly based on real monthly growth trends.

TOOLS

- **SQL:** This was the primary tool used for data extraction and cleaning to track monthly vital signs like loan applications and cash flow.
- **Power BI:** This tool was used to create the interactive report and visualize human trends such as why people are borrowing and where they live.
- **Excel:** This tool was used for initial data documentation and requirement gathering to ensure the project goals were clearly defined.